$$P_A(X) = \det(A - XI_n) = \begin{vmatrix} a_{11}-X & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22}-X & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn}-X \end{vmatrix}$$

$A = (a_{ij})$

$\text{grad } P_A = n$

rădăcinile polinomului caracteristic P_A = valorile proprii ale matricei A

$$P_A(X) = (-1)^n X^n + (-1)^{n-1} \operatorname{tr}(A) X^{n-1} + \dots + \det(A)$$

$X = 0 \qquad P_A(0) = \det(A)$

Obs $\lambda = 0 \in \sigma(A) \iff P_A(0) = 0 \iff$ spectrul matricei A

$\iff \det A = 0 \iff A$ necinductibilă

Thm. (Hamilton - Cayley) $P_A(A) = O_n$

$n = 2 \qquad A \in \mathcal{M}_2(\mathbb{R}) \qquad A^2 - \mathrm{Tr}(A)\,A + \det(A)\,I_2 = O_2$

$n = 3 \Rightarrow P_A(x) = \dots$

<u>exemplu</u> $\qquad A = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 2 & -4 \\ 0 & 0 & 3 \end{pmatrix}$

$$P_A(x) = \begin{vmatrix} 1-x & -1 & 2 \\ 0 & 2-x & -4 \\ 0 & 0 & 3-x \end{vmatrix} = (1-x)(2-x)(3-x) =$$

$$= -x^3 + 6x^2 - 11x + 6$$

$$\Rightarrow -A^3 + 6A^2 - 11A + 6 \cdot I_3 = O_3$$

$$I_3 = \frac{1}{6}\left(A^3 - 6A^2 + 11A\right)$$

$$= A \cdot \underbrace{\frac{1}{6}\left(A^2 - 6A + 11 I_3\right)}_{A^{-1}}$$

$\Rightarrow \quad A^{-1} \in Sp\{I_3, A, A^2\}$

$A^3 = 6A^2 - 11A + 6I_3 \in Sp\{I_3, A, A^2\}$

$A^4 = 6A^3 - 11A^2 + 6A =$

$= 6(6A^2 - 11A + 6I_3) - 11A^2 + 6A$

$A^4 \in Sp\{I_3, A, A^2\}$

$\Rightarrow \boxed{A^k \in Sp\{I_3, A, A^2\}} \subseteq M_3(\mathbb{R})$

$\forall\ k \in \mathbb{N} \quad (\ k \in \mathbb{Z}$ dacă A inversabilă$)$

Temă 1) $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ $\dim_{\mathbb{R}} Sp\{I_3, A, A^2\}$?

2) $A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ $\dim_{\mathbb{R}} Sp\{I_3, A, A^2\}$?

3) $A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = 2I_3$

$\dim_{\mathbb{R}} Sp\{I_3, A, A^2\} = 1$

$A \in \mathcal{M}_n(\mathbb{R})$ $\quad$ $\underline{P_A(x) \in \mathbb{R}[x]}$

$\Downarrow \quad \mathbb{R} \subseteq \mathbb{C}$

$A \in \mathcal{M}_n(\mathbb{C})$, $\lambda \in \mathbb{C}$, $v \in \mathbb{C}^n$

$\bar{\lambda} \in \mathbb{C}$, $\bar{v} \in \mathbb{C}^n$

$Av = \lambda v \Rightarrow A\bar{v} = \bar{\lambda}\bar{v}$

esempio $\quad A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ $\quad \sigma(A) = ?$ reali per.

$A \in \mathcal{M}_2(\mathbb{R})$ $\quad P_A(x) = x^2 - 0 \cdot x + 1 = x^2 + 1$

$\lambda = i$ $\quad \bar{\lambda} = -i$

$\sigma(A) = \{i, -i\}$

$\lambda = i \qquad v = \begin{pmatrix} z \\ w \end{pmatrix} \qquad Av = \lambda v$

$\Rightarrow \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix} = i \begin{pmatrix} z \\ w \end{pmatrix}$

$\Rightarrow \begin{cases} -w = iz \;/ : i \\ z = iw \end{cases}$

$V_\lambda = \left\{ \begin{pmatrix} iw \\ w \end{pmatrix} \mid w \in \mathbb{C} \right\} = Sp\left\{ \begin{pmatrix} i \\ 1 \end{pmatrix} \right\}$

$\dim_{\mathbb{C}} V_\lambda = 1$

$\overline{\lambda} = -i \qquad V_{\overline{\lambda}} = Sp\left\{ \begin{pmatrix} -i \\ 1 \end{pmatrix} \right\}$

$\dim_{\mathbb{C}} V_{\overline{\lambda}} = 1$

Thm. fundamentală a algebrei:

$\forall P \in \mathbb{C}[X] \Rightarrow$ există cel puțin o rădăcină $\lambda \in \mathbb{C}$ pt P

$P(X) = a(-1) \prod_{i=1}^{p} (X - \lambda_i)^{\alpha_i}$

($\Leftrightarrow$ P are toate rădăcinile în $\mathbb{C}$)

$\mathbb{C}$ cmp algebric inchis

$\mathbb{Q}$ $x^2 - 2$ — Z_p

$\cap$

$\mathbb{R}$ $x^2 + 1$ — $Z_2 = \{0, 1\}$ XOR AND

$\cap$

$\mathbb{C}$ $\equiv ax + b$

$x + b \quad -b = b$

$ax^2 + bx + c$

$x^2 \quad 0, 0$

$x^2 + x \quad 0, 1$

$(x+1)^2 = x^2 + 1 \quad 1, 1$

$x^2 + x + 1$ 

$Z_2 \subseteq K$

4

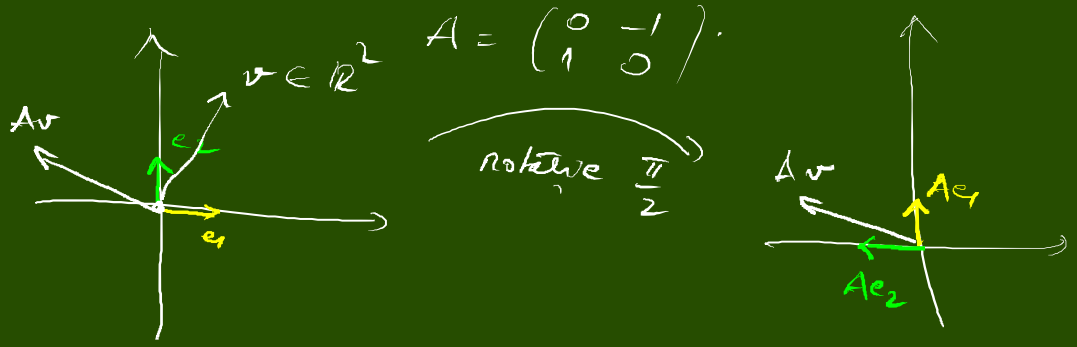
$\text{card}(K) = 4$

$16 \qquad 4$

$K \times K \longrightarrow K$ $\quad 4^{16}$ operatii posibile

$C^2_{416} + C^1_{416}$ perechi de operatii

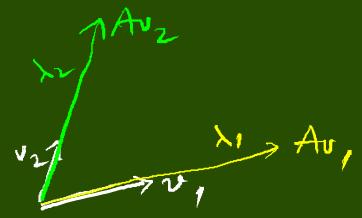




$$Ae_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = e_2$$

$$Ae_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} = -e_1$$

$Av = \lambda v$ $\quad$ v, Av collineari



$\lambda_1 \neq \lambda_2$

v_1, v_2 per lo collineari?

Dc. $v_2 = \alpha v_1$

$$Av_2 = \lambda_2 v_2 = \lambda_2 \alpha v_1$$

$$Av_2 = A(\alpha v_1) = \alpha \lambda_1 v_1$$

$\Rightarrow \alpha(\lambda_1 - \lambda_2)v_1 = 0 \Rightarrow \alpha = 0 \Rightarrow v_2 = 0$

$\neq 0 \quad \neq 0$ falso

$\Rightarrow v_1, v_2$ no colineales

$\mathbb{R}^3$ $\quad \mathcal{B}_1 = \{e_1, e_2\}$ $\quad \mathcal{B}_2 = \{e_3, \begin{pmatrix} 1 \\ 0 \end{pmatrix}\}$

$\mathcal{B}_1 \cup \mathcal{B}_2 = \{e_1, e_2, e_3, \begin{pmatrix} 1 \\ 0 \end{pmatrix}\}$

linealmente dependiente

$A = P \cdot D \cdot P^{-1}$

$D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$

$A = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 2 & -4 \\ 0 & 0 & 3 \end{pmatrix}$

$\begin{cases} \lambda_1 = 1 & v_1 \neq 0 \\ \lambda_2 = 2 & v_2 \neq 0 \\ \lambda_3 = 3 & v_3 \neq 0 \end{cases}$

$\Rightarrow \quad A = P \cdot D \cdot P^{-1}$

$P = (v_1 | v_2 | v_3) \qquad D = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{pmatrix}$

$(\dots)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 1 \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$\lambda_1 = 1 \qquad Av_1 = \lambda_1 v_1 \qquad v_1 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$\Rightarrow \begin{cases} x - y + 2z = x \\ 2y - 4z = y \Rightarrow y = 4z \\ 3z = z \end{cases}$

$\Rightarrow \quad z = 0 \,, \; y = 0 \,, \; x \in \mathbb{R}$

$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad x = 1$

$\lambda_2 = 2 \qquad Av_2 = \lambda_2 v_2 \qquad v_2 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$\begin{cases} x - y + 2z = 2x \qquad x = -y \\ 2y - 4z = 2y \\ 3z = 2z \quad \Rightarrow z = 0 \end{cases}$

$$v_2 = \begin{pmatrix} x \\ -x \\ 0 \end{pmatrix} \qquad x \in \mathbb{R}$$

$$x = 1 \Rightarrow v_2 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

$\lambda_3 = 3$

$$\begin{cases} x - y + 2z = 3x \\ 2y - 4z = 3y \\ 3z = 3z \end{cases} \qquad y = -4z$$

$$4z + 2z = 2x \qquad x = 3z$$

$$v_3 = \begin{pmatrix} 3z \\ -4z \\ 2 \end{pmatrix} \qquad z \in \mathbb{R}$$

$z = 1 \qquad v_3 = \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$

$$\Rightarrow \quad A = P \cdot D \cdot P^T$$

$$P = \begin{pmatrix} 1 & 1 & 3 \\ 0 & -1 & -4 \\ 0 & 0 & 1 \end{pmatrix} \qquad D = \begin{pmatrix} 1 & & \\ & 2 & \\ & & 3 \end{pmatrix}$$

$\lambda \in \sigma(A)$

$$1 \leq g(\lambda) \leq a(\lambda) \leq n$$

$A, B \in \mathcal{M}_n(k)$

$A = P \cdot B \cdot P^{-1}$ ($A \sim B$ matrice asemănătoare)

$\Rightarrow P_A(x) = P_B(x)$ $\qquad B = \begin{pmatrix} \lambda & \cdots & \lambda & | & * \\ & & & | & \\ 0 & & & | & \end{pmatrix}$

$= (x-\lambda)^{g(\lambda)} \cdot Q(x)$

$g(\lambda) \leq a(\lambda)$

Algoritm de diagonalizare $A \in \mathcal{M}_n(k)$

I Calc $P_A(x)$ și rădăcinile $\lambda \in \sigma(A)$ cu ordinele de multiplicitate $a(\lambda)$

II $\forall \lambda \in \sigma(A)$ determinăm V_λ $\xleftarrow{\text{bază } B_\lambda}$ $\xleftarrow{\text{dim } V_\lambda =}$

$g(\lambda)$

III $\forall \lambda \in \sigma(A)$, $g(\lambda) = a(\lambda)$? $\Leftarrow$ NU $\equiv$ STOP

DA $\Rightarrow$ IV

IV $\{v_1 \dots v_n\} = \bigcup B_\lambda$ bază în k^n $\lambda \in \sigma(A)$

Jordan ($n=2$)

$P = (v_1 | \dots | v_n)$ $\quad D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$

$A = P \cdot D \cdot P^{-1}$

(verificare : $AP = PD$)

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exemplu : $A = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$, $\tilde{A} = \begin{pmatrix} -2 & 1 & -1 \\ 0 & -2 & -5 \\ 0 & 0 & 3 \end{pmatrix}$

temă

$P_A(x) = \begin{vmatrix} -2-x & 1 & 1 \\ 1 & -2-x & 1 \\ 1 & 1 & -2-x \end{vmatrix} =$

$$= \begin{vmatrix} -x & 1 & 1 \\ -x & -2-x & 1 \\ -x & 1 & -2-x \end{vmatrix} = (-x) \begin{vmatrix} 1 & 1 & 1 \\ 1 & -2-x & 1 \\ 1 & 1 & -2-x \end{vmatrix}$$

$$= (-x) \begin{vmatrix} 1 & 1 & 1 \\ 0 & -3-x & 0 \\ 0 & 0 & -3-x \end{vmatrix} =$$

$$= (-x)(-3-x)^2 = -x(x+3)^2$$

$$= -(x-0)^{1}(x+3)^{2}$$

$\lambda_1 = 0 \quad a(\lambda_1) = 1$

$\lambda_2 = -3 \quad a(\lambda_2) = 2$

$\lambda_1 = 0 \quad V_{\lambda_1} = \{ v \in \mathbb{R}^3 \mid Av = \lambda_1 v \}$

$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad Av = \lambda_1 v \Rightarrow A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$\lambda = 0$ val. pr. $\Rightarrow V_\lambda = \ker(A)$

$$\begin{cases} -2x + y + z = 0 \\ x - 2y + z = 0 \\ x + y - 2z = 0 \end{cases} \Rightarrow x = y \Big|_{\Rightarrow x = z}$$

$\Rightarrow \quad x = y = z$

$$V_{\lambda_1} = \left\{ \begin{pmatrix} x \\ x \\ x \end{pmatrix} \mid x \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\} = v_1$$

$$1 \leq g(\lambda_1) = \dim_{\mathbb{R}} V_{\lambda_1} = 1 \leq a(\lambda_1) = 1$$

$\lambda_2 = -3 \qquad V_{\lambda_2} = \{ v \in \mathbb{R}^3 \mid Av = \lambda_2 v \}$

$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \qquad Av = \lambda_2 v \Rightarrow Av = -3v$

$$A\begin{pmatrix} x \\ y \\ z \end{pmatrix} = -3\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{cases} -2x + y + z = -3x \\ x - 2y + z = -3y \\ x + y - 2z = -3z \end{cases}$$

$\Rightarrow \quad x+y+z=0 \quad \Rightarrow \quad z=-x-y$

$$V_{\lambda_2} = \left\{ \begin{pmatrix} x \\ y \\ -x-y \end{pmatrix} \mid x,y \in \mathbb{R} \right\} = Sp\left\{ \underbrace{\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}}_{v_2}, \underbrace{\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}}_{v_3} \right\}$$

$$1 \le g(\lambda_2) = \dim_{\mathbb{R}} V_{\lambda_2} = 2 \le a(\lambda_2) = 2$$

Deci λ_1 : $g(\lambda_1) = a(\lambda_1) = 1$

λ_2 : $g(\lambda_2) = a(\lambda_2) = 2$

$\Rightarrow$ A diagonalizabilă $\quad A = P \cdot D \cdot P^{-1}$

$$P = (v_1 \mid v_2 \mid v_3) = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & -1 \end{pmatrix}$$

$$D = \begin{pmatrix} 0 & & \\ & -3 & \\ & & -3 \end{pmatrix}$$

$$A = \begin{pmatrix} 5 & 1 & 2 \\ 0 & 5 & 4 \\ 0 & 0 & 5 \end{pmatrix} \qquad P_A(X) = -(X-5)^3$$

$\lambda = 5 \quad a(\lambda) = 3$

dc $A = P \cdot D \cdot P^{-1}$ $\qquad D = \begin{pmatrix} 5 & & \\ & 5 & \\ & & 5 \end{pmatrix} = 5I_3$

$= P \cdot 5I_3 \cdot P^{-1}$

$= 5 \cdot I_3$, fals $\qquad \begin{cases} A = \lambda I_n \\ A \neq \lambda I_n \end{cases}$

$\Rightarrow A$ nu e diagonalizabilă.
